

UNIT-I: LOGIC AND PROPOSITIONS

1.1 Introduction to Mathematical Logic

Logic is the science of correct reasoning. It provides the foundation for mathematics, computer science, artificial intelligence, digital circuit design, database systems, and many other disciplines.

In mathematics, logic is used to determine whether an argument is valid and whether a statement is true or false.

The word "logic" is derived from the Greek word *logos*, meaning reason or thought.

Logic helps us:

- Analyze mathematical statements.
- Construct mathematical proofs.
- Verify computer programs.
- Design digital circuits.
- Develop algorithms.

Every mathematical theorem and proof is based on logical reasoning.

1.2 Propositions (Statements)

Definition

A proposition or statement is a declarative sentence that has a definite truth value, either True (T) or False (F), but not both simultaneously.

In logic, only propositions can be analyzed for truth or falsity.

Examples of Propositions

Example 1

7 is a prime number.

Truth Value: True

Reason:

The only positive divisors of 7 are 1 and 7.

Example 2

12 is an odd number.

Truth Value: False

Reason:

12 is divisible by 2 and therefore even.

Example 3

The Earth revolves around the Sun.

Truth Value: True

Example 4

$2 + 3 = 8$

Truth Value: False

Example 5

Every square has four sides.

Truth Value: True

1.3 Non-Propositions

A sentence that does not have a definite truth value is not a proposition.

Examples

Question

What is your name?

This is not a proposition because it asks for information.

Command

Close the door.

This is not a proposition because commands cannot be true or false.

Exclamation

What a beautiful day!

This expresses emotion rather than a factual statement.

Open Sentence

$$x + 3 = 7$$

This is not a proposition because its truth depends on the value of x .

When $x = 4$, the statement is true.

When $x = 5$, the statement is false.

1.4 Types of Propositions

Propositions may be classified as:

Simple (Atomic) Proposition

A proposition that cannot be broken into smaller propositions.

Examples:

- 5 is prime.
- Today is Monday.

- Odisha is a state of India.
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Compound Proposition

A proposition formed by combining two or more propositions using logical connectives.

Examples:

- It is raining and I carry an umbrella.
 - Ram passed Mathematics or Physics.
 - If it rains, then roads become wet.
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1.5 Truth Values

Every proposition has exactly one truth value.

Truth values are:

T = True

F = False

Example:

Proposition	Truth Value
5 is prime	T
9 is prime	F
10 is even	T
11 is even	F

1.6 Compound Propositions

Let

p : It is raining.

q : I carry an umbrella.

Using logical connectives we can construct:

- $p \wedge q$

- $p \vee q$

- $\neg p$

- $p \rightarrow q$

- $p \leftrightarrow q$

These are called compound propositions.

1.7 Logical Connectives

Logical connectives are symbols used to combine propositions.

The fundamental connectives are:

1. Negation
 2. Conjunction
 3. Disjunction
 4. Exclusive OR
 5. Implication
 6. Biconditional
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1.8 Negation

Definition

The negation of a proposition p is denoted by $\neg p$.

It means "not p ."

Example 1

p : Today is Monday.

$\neg p$: Today is not Monday.

Example 2

p : 7 is prime.

$\neg p$: 7 is not prime.

Since p is true,

$\neg p$ is false.

Truth Table

p	$\neg p$
T	F
F	T

Important Note

Negation reverses the truth value.

True becomes False.

False becomes True.

1.9 Conjunction (AND)

Definition

The conjunction of p and q is written as

$p \wedge q$

and read as

" p and q ."

A conjunction is true only when both propositions are true.

Example

p : Ram passed Mathematics.

q : Ram passed Physics.

Then

$p \wedge q$

means

"Ram passed Mathematics and Physics."

Truth Table

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Real-Life Example

A scholarship requires:

- CGPA above 8.0

AND

- Attendance above 75%

A student receives the scholarship only when both conditions are satisfied.

1.10 Disjunction (OR)

Definition

The disjunction of p and q is written as

$$p \vee q$$

and read as

" p or q ."

In mathematics, OR means inclusive OR.

Example

A student may submit the assignment:

- Online

OR

- Offline

OR

- Both

Truth Table

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Common Mistake

Students often think OR means exactly one.

In mathematics, OR allows both statements to be true.

1.11 Exclusive OR (XOR)

Definition

Exclusive OR means exactly one proposition is true.

Symbol:

$$p \oplus q$$

Example

A switch can be:

- ON

or

- OFF

but not both.

Truth Table

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

1.12 Implication

Definition

An implication has the form

$$p \rightarrow q$$

and is read as:

"If p, then q."

where

p = hypothesis

q = conclusion

Example 1

If a number is divisible by 4, then it is even.

Example 2

If a student studies regularly, then the student performs well.

Truth Table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Important Observation

An implication is false only when:

p is true and q is false.

This is the most important property of implication.

1.13 Biconditional

Definition

The biconditional is written as

$$p \leftrightarrow q$$

and read as

" p if and only if q ."

Example

An integer n is even if and only if $n = 2k$ for some integer k .

Truth Table

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

1.14 Tautologies, Contradictions, and Contingencies

Tautology

A compound proposition that is always true.

Example:

$$p \vee \neg p$$

This is called the Law of Excluded Middle.

Contradiction

A compound proposition that is always false.

Example:

$$p \wedge \neg p$$

This is called the Law of Contradiction.

Contingency

A proposition that is neither always true nor always false.

Example:

$$p \wedge q$$

Its truth depends on the truth values of p and q .

Summary

In this chapter we studied:

- Logic and reasoning
- Propositions and non-propositions
- Simple and compound propositions
- Truth values
- Negation

- Conjunction
- Disjunction
- Exclusive OR
- Implication
- Biconditional
- Tautologies, contradictions, and contingencies

These concepts form the foundation for all subsequent topics in Discrete Mathematics.